



**BAULKHAM HILLS
HIGH
SCHOOL**

2018

**YEAR 12 TRIAL
HIGHER
SCHOOL
CERTIFICATE
EXAMINATION**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Write your NESAs# and Teacher's name on your answer booklet
- Board-approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for careless or badly arranged work

Total marks: 70

Section I – 10 marks (pages 2 – 5)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 6 – 12)

- Attempt Questions 11 – 14
- Allow about 1 hour 45 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

- 1 If the acute angle between the lines $y = 2x + 5$ and $mx - y - 3 = 0$ is 45° , the possible values of m are

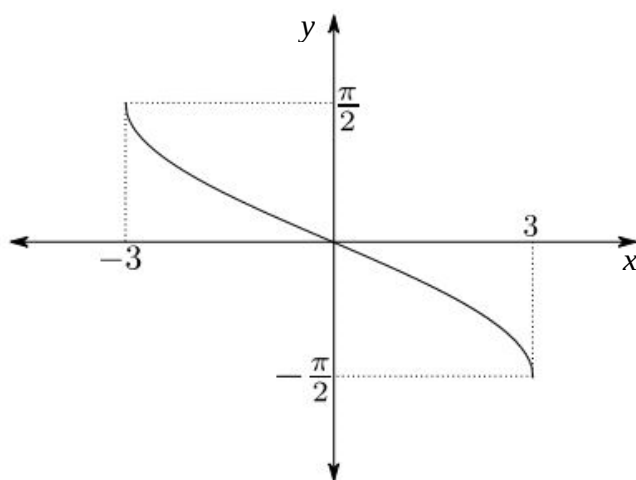
(A) $\frac{1}{3}$ and -3

(B) $\frac{1}{2}$ and -2

(C) $-\frac{1}{2}$ and 2

(D) $-\frac{1}{3}$ and 3

2



The correct function for the graph above is

(A) $y = -\sin^{-1}\left(\frac{x}{3}\right)$

(B) $y = -\sin^{-1} 3x$

(C) $y = \frac{\pi}{2} - \cos^{-1}\left(\frac{x}{3}\right)$

(D) $y = \frac{\pi}{2} - \cos^{-1} 3x$

- 3 When the polynomial $P(x)$ is divided by $(x + 3)(x - 4)$ the remainder is $(3x + 2)$.

What is the remainder when $P(x)$ is divided by $(x - 4)$?

(A) -10

(B) -7

(C) 11

(D) 14

- 4 A particle is moving along the x -axis such that its velocity, v , at position, x , is given by $v = \sqrt{10x - x^2}$.

What is the acceleration of the particle when $x = 1$?

(A) $\frac{4}{3}$

(B) $\frac{8}{3}$

(C) 3

(D) 4

- 5 Which of the following is an asymptote of the curve $y = \frac{x^2 - 1}{x}$?

(A) $y = x$

(B) $y = 0$

(C) $y = 1$

(D) $x = 1$

6 Which of the following are true for all real values of x ?

I $\sin\left(\frac{\pi}{2} + x\right) = \cos\left(\frac{\pi}{2} - x\right)$

II $2 + 2\sin x - \cos^2 x \geq 0$

III $\sin\left(x + \frac{3\pi}{2}\right) = \cos(\pi - x)$

IV $\sin x \cos x \leq \frac{1}{4}$

(A) **I** and **II**

(B) **II** and **III**

(C) **II** and **IV**

(D) **III** and **IV**

7 The letters of the word **PERSEVERE** are arranged in a row. The number of different arrangements that are possible if all of the four **E**'s remain together are

(A) $6!$

(B) $\frac{6!}{2!}$

(C) $\frac{9!}{2!}$

(D) $\frac{9!}{4!2!}$

8 If a focal chord of the parabola $x^2 = 4ay$ cuts the parabola at two distinct points (x_1, y_1) and (x_2, y_2) , then;

(A) $x_1 x_2 = a^2$

(B) $y_1 y_2 = a^2$

(C) $x_1(x_2)^2 = a^2$

(D) $y_1(y_2)^2 = a^2$

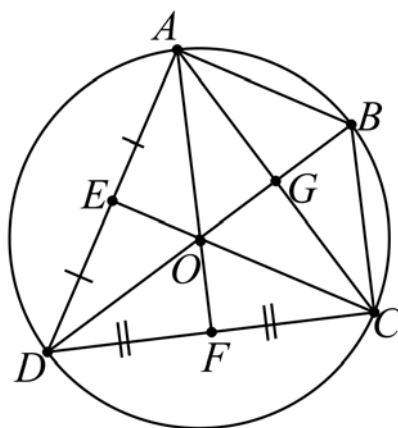
- 9 Consider two functions

$$f(x) = a - x^2$$

$$g(x) = x^4 - a$$

For precisely which values of $a > 0$ is the area of the region bounded by the x -axis and the curve $y = f(x)$ bigger than the area of the region bounded by the x -axis and the curve $y = g(x)$?

- (A) $a > 1$
- (B) $a > \frac{6}{5}$
- (C) $a > \left(\frac{4}{3}\right)^{\frac{3}{2}}$
- (D) $a > \left(\frac{6}{5}\right)^4$
- 10 A, B, C and D are concyclic points on a circle centre O . E, F and G are points on the chords AD, CD and AC respectively, such that AF, CE and DG are concurrent at O , and D, O, G and B are collinear. $AE = DE$ and $CF = DF$.



Which of the following statements is **NOT** true?

- (A) $\angle BAD = 90^\circ$
- (B) $\angle AGD = 90^\circ$
- (C) $ABCO$ is a cyclic quadrilateral
- (D) ΔABO is similar to ΔACD

END OF SECTION I

Section II

90 marks

Attempt Questions 11 – 14

Allow about 1 hour 45 minutes for this section

Answer each question on the appropriate page in the answer booklet. Extra paper is available, write your NESAS# on any extra paper you use.

In Questions 11 to 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start on the page labelled Question 11 in your answer booklet

Marks

- | | |
|---|---|
| (a) Find $\int \cos^2 x \, dx$ | 2 |
| (b) Solve $\frac{2x}{5-x} \geq 1$ | 3 |
| (c) Differentiate $\sin^{-1}(x^2)$ | 2 |
| (d) Express $15\cos x - 8\sin x$ in the form $A\cos(x + \alpha)$, where $0 \leq \alpha \leq \frac{\pi}{2}$ | 2 |
| (e) You are given that in the expansion of $(a + bx)^5$, the constant term is 32 and the coefficient of x^3 is -1080 .
Find the values of a and b . | 3 |
| (f) A particle moves so that its distance, x centimetres, from a fixed point O at time, t seconds, is $x = 4\sin 2t$. | |
| (i) Show that the particle is moving in Simple Harmonic Motion | 2 |
| (ii) What is the frequency of the particle's motion, in oscillations per second? | 1 |

Question 12 (15 marks) Start on the page labelled Question 12 in your answer booklet

(a) Evaluate $\lim_{x \rightarrow 2} \frac{\sin(2-x)}{(x-2)(x+3)}$ 2

(b) Celeste and Michelle are playing a table tennis match. The winner of the match is the first player to win three games.

The probability that Celeste wins a game is 0.55, games cannot be drawn.

Find, correct to two decimal places, the probability that

(i) Celeste wins the match in three games. 1

(ii) Celeste wins the match. 2

(c) Use mathematical induction to prove that 3

$$\frac{2}{1 \times 2 \times 3} + \frac{2}{2 \times 3 \times 4} + \frac{2}{3 \times 4 \times 5} + \dots + \frac{2}{n(n+1)(n+2)} = \frac{1}{2} - \frac{1}{(n+1)(n+2)}$$

for all integers $n \geq 1$

(d) Using the expansion of $(1+x)^n$, find the value of

(i) $\binom{n}{0} + 2\binom{n}{1} + 4\binom{n}{2} + \dots + 2^n\binom{n}{n}$ 1

(ii) $\binom{n}{1} - 2\binom{n}{2} + 3\binom{n}{3} - \dots + (-1)^{n-1}n\binom{n}{n}$ 2

Question 12 continues on page 8

Question 12 (continued)

- (e) It is known that a root of the equation $e^x - 2x^2 = 0$ exists in the interval $0.6 < x < 2.4$

- (i) Use one application of Newton's method to find a three decimal place approximation to a root of the equation $e^x - 2x^2 = 0$, using $x_0 = 1.5$ as a first approximation. 2

- (ii) Copy and complete, correct to two decimal places, the following table of values for $P(x) = e^x - 2x^2$ 1

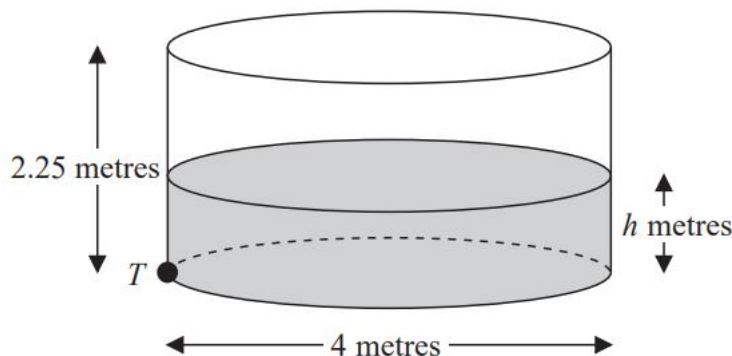
x	2.0	2.1	2.2	2.3	2.4
$P(x)$					

- (iii) Hence, without further calculation, explain why $x_0 = 2.3$ would have been a less suitable first approximation for the root of $P(x) = 0$ that lies in the interval $0.6 < x < 2.4$ 1

End of Question 12

Question 13 (15 marks) Start on the page labelled Question 13 in your answer booklet

- (a) A cylindrical tank has diameter 4 metres and height 2.25 metres. Water is flowing into the tank at a rate of $\frac{2\pi}{5} \text{ m}^3/\text{min}$.
 There is a tap at a point T at the base of the tank. When the tap is opened, water leaves the tank at a rate of $\frac{\pi}{5}\sqrt{h} \text{ m}^3/\text{min}$, where h is the height of the water in metres.



- (i) Show that at time t minutes after the tap has opened, the volume of water in the tank satisfies the differential equation 1

$$\frac{dV}{dt} = \frac{\pi(2 - \sqrt{h})}{5}$$

- (ii) Show that at time t minutes after the tap has opened, the height of the water in the tank satisfies the differential equation 2

$$\frac{dh}{dt} = \frac{2 - \sqrt{h}}{20}$$

- (iii) When the tap is opened the height of the water is 0.16 metres. The time taken to fill the tank to a height of 2.25 metres can be calculated using 3

$$t = \int_{0.16}^{2.25} \frac{20}{2 - \sqrt{h}} dh \quad (\text{Do NOT prove this})$$

Using the substitution $h = (2 - x)^2$, where $0 < x < 2$, find the time taken to fill the tank, correct to the nearest minute.

Question 13 continues on page 10

Question 13 (continued)

- (b) Two hundred rabbits in a region with an estimated population of 200 000 rabbits have a highly contagious disease.

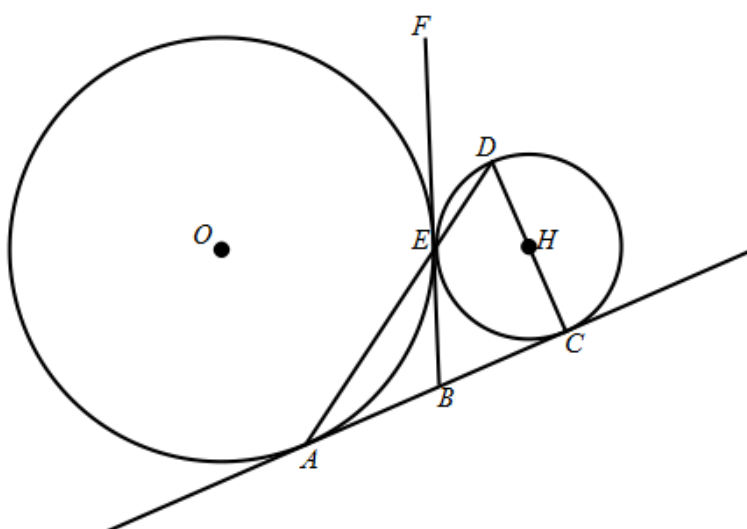
The disease is known to spread at the weekly rate of 1% of the remaining healthy rabbits such that

$$\frac{dP}{dt} = 0.01(200\,000 - P)$$

where P is the number of infected rabbits after t weeks.

- (i) Show that $P = 200\,000 - 199\,800e^{-0.01t}$ satisfies both the differential equation and the initial conditions. 2
- (ii) How many days does it take for half of the rabbit population to become infected? 3

- (c) 4



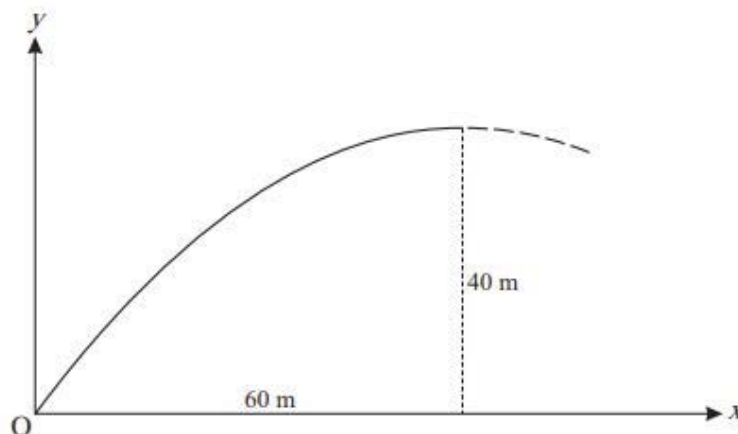
In the diagram, two circles with centres O and H touch externally at E .
The common tangent at E meets another common tangent AC at B .
 CD is a diameter of the smaller circle.

Copy the diagram into your answer booklet and prove that A , E and D are collinear.

End of Question 13

Question 14 (15 marks) Start on the page labelled Question 14 in your answer booklet

- (a) A small firework is fired at ground level with initial speed $V \text{ ms}^{-1}$ at an angle of θ to the horizontal. The highest point reached by the firework is at a horizontal distance of 60 metres from the point of projection and a vertical distance of 40 metres above the ground.



Neglecting the effects of air resistance, the equations describing the motions of the firework are

$$x = Vt \cos \theta$$

$$y = Vt \sin \theta - 4.9t^2$$

where t is the time in seconds after the firework is projected. Do NOT prove these equations.

It is known that the initial horizontal velocity of the firework is 21 ms^{-1}

- (i) Calculate the time for the firework to reach its highest point, correct to two decimal places 2
- (ii) Show that the initial vertical velocity is 28 ms^{-1} 2

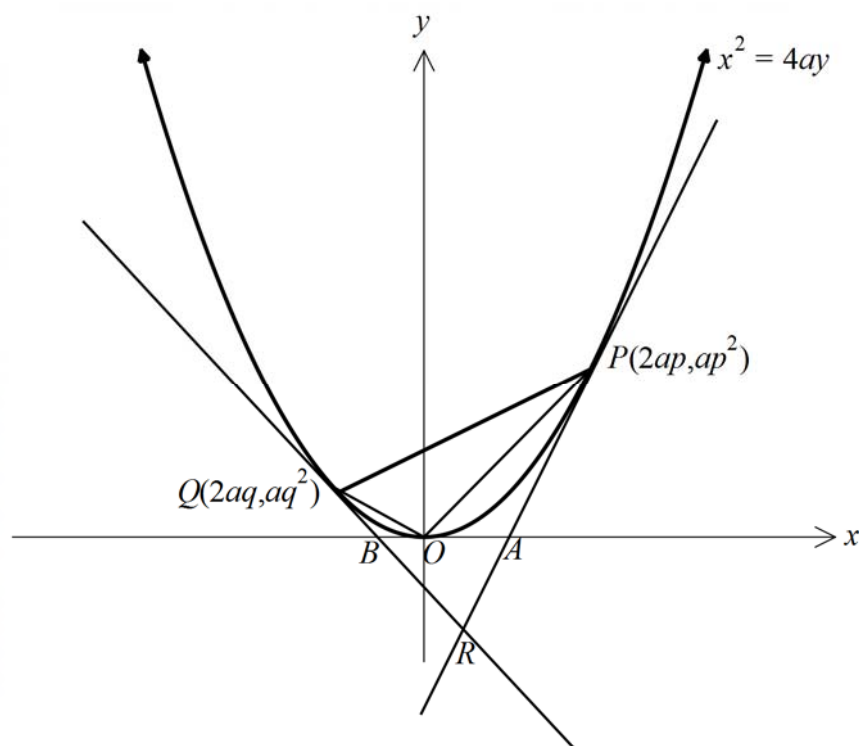
When the firework is at its highest point it explodes into several parts. Two of these parts initially continue to travel horizontally, one with the original horizontal speed of 21 ms^{-1} and the other with a quarter of this speed.

- (iii) State why the two parts are always at the same height as one another above the ground 1
- (iv) Find the distance between the two parts of the firework when they hit the ground, correct to the nearest metre. 2

Question 14 continues on page 12

Question 14 (continued)

- (b) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$, where $p > 0$ and $q < 0$, and $|p| > |q|$, lie on the parabola $x^2 = 4ay$.

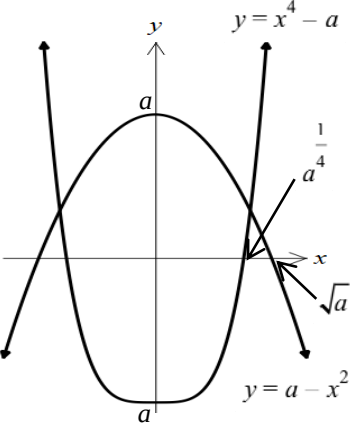
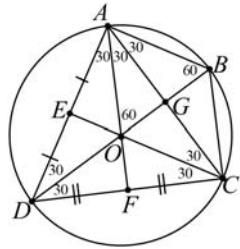


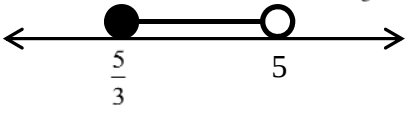
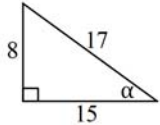
- | | | |
|-------|---|---|
| (i) | Write down the equations of the tangents to the parabola at P and Q | 1 |
| (ii) | The tangents to the parabola at P and Q meet at R .
Show that R has coordinates $\{a(p + q), apq\}$ | 2 |
| (iii) | The tangents at P and Q meet the x -axis at A and B respectively.
Show that the area of $\triangle ABR$ is $\frac{1}{2} a^2 pq(q - p)$ | 2 |
| (iv) | Prove that the area of $\triangle OPQ$ is twice the area of $\triangle ABR$ | 3 |

End of paper

BAULKHAM HILLS HIGH SCHOOL
YEAR 12 EXTENSION 1 TRIAL 2018 SOLUTIONS

Solution	Marks	Comments
SECTION I		
1. A – $y = 2x + 5 \Rightarrow m_1 = 2$ $mx - y - 3 = 0 \Rightarrow m_2 = m$ $\tan 45^\circ = \left \frac{2 - m}{1 + 2m} \right $ $1 = \left \frac{2 - m}{1 + 2m} \right $ $ 2 - m = 1 + 2m $ $2 - m = 1 + 2m \quad \text{or} \quad -(2 - m) = 1 + 2m$ $-3m = -1 \quad \quad \quad m - 2 = 1 + 2m$ $m = \frac{1}{3} \quad \quad \quad m = -3$	1	
2. A – Domain : $-1 \leq ax \leq 1$ Curve is either $\sin^{-1}f(x)$ flipped upside down $-\frac{1}{a} \leq x \leq \frac{1}{a}$ i.e. $-\sin^{-1}f(x)$ or $\therefore \frac{1}{a} = 3$ $\cos^{-1}f(x)$ shifted down $\frac{\pi}{2}$ i.e. $\cos^{-1}f(x) - \frac{\pi}{2}$ $a = \frac{1}{3}$ $\therefore$ the correct function is $y = -\sin^{-1}\left(\frac{x}{3}\right)$	1	
3. D – $P(x) = (x+3)(x-4)Q(x) + (3x+2)$ $P(4) = 0 + 3(4) + 2$ $= 14$	1	
4. D – $\ddot{x} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$ when $x = 1$; $\ddot{x} = 5 - 1$ $= 4$ $= \frac{d}{dx}\left(5x - \frac{1}{2}x^2\right)$ $= 5 - x$	1	
5. A - $y = \frac{x^2 - 1}{x}$ asymptotes are $y = x$ and $x = 0$ $= x - \frac{1}{x}$	1	
6. B - I: Let $x = 0$, $\sin\frac{\pi}{2} = \cos\frac{\pi}{2}$ ✗ II: $2 + 2\sin x - \cos^2 x = 2 + 2\sin x - 1 + \sin^2 x$ ✓ $= \sin^2 x + 2\sin x + 1$ $= (\sin x + 1)^2 \geq 0$ III: $\sin\left(x + \frac{3\pi}{2}\right) = \cos\left\{\frac{\pi}{2} - \left(x + \frac{3\pi}{2}\right)\right\}$ ✓ $= \cos(\pi - x)$ IV: $\sin x \cos x = \frac{1}{2}\sin 2x$ ✗ Thus the two that are correct are II and III $\leq \frac{1}{2}$	1	
7. B – The four E's are now treated as one letter, so the question becomes how many arrangements of PRSVR(EEEE) Ways $= \frac{6!}{2!}$	1	

Solution	Marks	Comments
8. B - $\frac{m_{SP} = m_{SQ}}{y_1 - a = y_2 - a} = \frac{x_1}{x_2}$ $\frac{x_2 y_1 - a x_2}{x_1^2 x_2} = \frac{x_1 y_2 - a x_1}{x_1 x_2^2 - a x_1}$ $\frac{x_1 x_2 (x_1 - x_2)}{x_1 x_2} = -4a^2$ $(x_1 x_2)^2 = 16a^4$ $4ay_1 \times 4ay_2 = 16a^4$ $y_1 y_2 = a^2$	1	
9. D - $\int_0^{\sqrt{a}} a - x^2 dx > - \int_0^{\sqrt{a}} x^4 - a dx$ $\left[ax - \frac{1}{3}x^3 \right]_0^{\sqrt{a}} > - \left[\frac{1}{5}x^5 - ax \right]_0^{\sqrt{a}}$ $\frac{3}{2}a^{\frac{3}{2}} - \frac{1}{3}a^{\frac{3}{2}} > -\frac{1}{5}a^{\frac{5}{2}} + a^{\frac{5}{2}}$ $\frac{2}{3}a^{\frac{3}{2}} - \frac{4}{5}a^{\frac{5}{2}} > 0$ $a^{\frac{5}{2}} \left(\frac{2}{3}a^{-\frac{1}{2}} - \frac{4}{5} \right) > 0$ By definition $a^{\frac{5}{2}} > 0$ Thus $\frac{2}{3}a^{-\frac{1}{2}} - \frac{4}{5} > 0$ $a^{-\frac{1}{2}} > \frac{4}{5} \times \frac{3}{2}$ $a > \left(\frac{6}{5} \right)^4$ 	1	
10. C - $\angle BAD = 90^\circ$ ($\angle$ in a semicircle) $\angle AGD = 90^\circ$ AF $\perp$ CD, CE $\perp$ AD (perp from centre bisects chord) $\therefore$ DG $\perp$ AC (altitudes are concurrent)  altitudes, medians, perpendicular bisectors are the same lines $\therefore \Delta ACD$ is equilateral $\angle BOA = 60^\circ$ ($\angle$ at centre, " twice $\angle$ at circumference) so ΔABO also turns out to be equilateral Thus $\Delta ABO \parallel \Delta ACD$ $ABCO$ is NOT a cyclic quadrilateral as opposite $\angle$'s are NOT supplementary $\therefore$ answer is C	1	

SECTION II		
Solution	Marks	Comments
QUESTION 11		
11(a) $\int \cos^2 x \, dx = \frac{1}{2} \int 1 + \cos 2x \, dx$ $= \frac{1}{2} \left(x + \frac{1}{2} \sin 2x \right) + c$ $= \frac{x}{2} + \frac{1}{4} \sin 2x + c$	2	2 marks • Correct solution 1 mark • Find a correct relationship between $\cos^2 x$ and $\cos 2x$
11(b) $\frac{2x}{5-x} \geq 1$ $5-x \neq 0$ $x \neq 5$ $\frac{2x}{5-x} = 1$ $2x = 5-x$ $3x = 5$ $x = \frac{5}{3}$  $\frac{5}{3} \leq x < 5$	3	3 marks • Correct graphical solution on number line or algebraic solution, with correct working 2 marks • Bald answer • Identifies the two correct critical points via a correct method • Correct conclusion to their critical points obtained using a correct method 1 mark • Uses a correct method • Acknowledges a problem with the denominator. 0 marks • Solves like a normal equation, with no consideration of the denominator.
11(c) $f(x) = \sin^{-1}(x^2)$ $f'(x) = \frac{2x}{\sqrt{1-x^4}}$	2	2 marks • Correct solution 1 mark • obtains $\frac{g(x)}{\sqrt{1-x^4}}$ or equivalent merit
11(d)  $\alpha = \tan^{-1}\left(\frac{8}{15}\right)$ $= 0.4899573263...$ $15\cos x - 8\sin x$ $= 17\cos\left(x + \tan^{-1}\frac{8}{15}\right)$ OR $= 17\cos(x + 0.4899573263...)$	2	2 marks • Correct solution 1 mark • Finds A • establishes $\alpha = \tan^{-1}\frac{8}{15}$ <i>Note: no penalty for rounding, if it is clear how α has been established</i>
11(e) $a^5 = 32$ $a = 2$ $\left(\frac{5}{3}\right)a^2(bx)^3 = -1080x^3$ $(10)(4)b^3 = -1080$ $b^3 = -27$ $b = -3$	3	3 marks • Correct solution 2 marks • Finds b • Finds a and the term involving x^3 1 mark • Finds a • Finds the term involving x^3

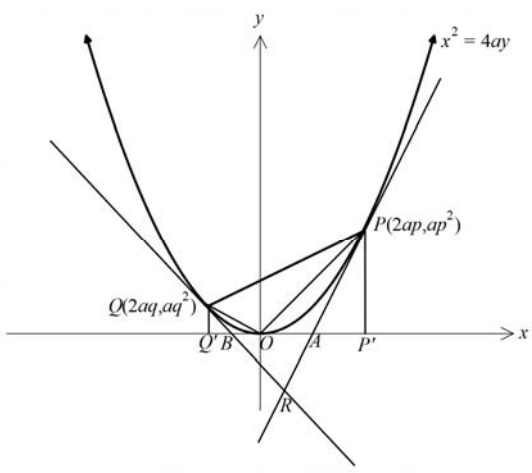
Solution		Marks	Comments
11(f) (i)	$x = 4\sin 2t$ $\dot{x} = 8\cos 2t$ $\ddot{x} = -16\sin 2t$ $= -4(4\sin 2t)$ $= -4x$ $\therefore \text{particle moves in SHM as } \ddot{x} = -n^2x$	2	2 marks • Correct solution 1 mark • Recognises the condition for a particle to move in SHM • Correctly obtains acceleration as a function of time by differentiation
11 (f) (ii)	$\dot{x} = -4x$ $\therefore n = 2$ $f = \frac{n}{2\pi}$ $= \frac{2}{2\pi}$ $= \frac{1}{\pi}$ $\therefore \text{frequency of the particle is } \frac{1}{\pi} \text{ oscillations/second (Hz)}$	1	1 mark • Correct answer
QUESTION 12			
12 (a)	$\lim_{x \rightarrow 2} \frac{\sin(2-x)}{(x-2)(x+3)} = \lim_{x \rightarrow 2} \frac{\sin(2-x)}{(2-x)} \times \frac{-1}{(x+3)}$ $= 1 \times -\frac{1}{5}$ $= -\frac{1}{5}$	2	2 marks • Correct solution 1 mark • Attempts to use the “small angle” theorem
12 (b) (i)	$P(\text{Celeste wins in 3 games}) = (0.55)^3$ $= 0.166375$ $= 0.17 \text{ (to 2 dp)}$	1	1 mark • Correct answer
12 (b) (ii)	In order for Celeste to win, she must win the last game and two others. Michelle could win 0, 1 or 2 games $P(\text{Celeste wins}) = 0.55 \left\{ \binom{2}{0} (0.55)^2 (0.45)^0 + \binom{3}{1} (0.55)^2 (0.45) + \binom{4}{2} (0.55)^2 (0.45)^2 \right\}$ $= 0.593126875...$ $= 0.59 \text{ (to 2 dp)}$	2	2 marks • Correct solution 1 mark • Establishes multiple situations where Celeste wins
12 (c)	When $n = 1$; $LHS = \frac{2}{1 \times 2 \times 3}$ $= \frac{2}{6}$ $= \frac{1}{3}$ $RHS = \frac{1}{2} - \frac{1}{2 \times 3}$ $= \frac{1}{2} - \frac{1}{6}$ $= \frac{1}{3}$ $\therefore LHS=RHS$ Hence the result is true for $n = 1$ Assume the result is true for $n = k$ $\text{i.e. } \frac{2}{1 \times 2 \times 3} + \frac{2}{2 \times 3 \times 4} + \frac{2}{3 \times 4 \times 5} + \dots + \frac{2}{k(k+1)(k+2)} = \frac{1}{2} - \frac{1}{(k+1)(k+2)}$ Prove the result is true for $n = k + 1$ $\text{i.e. } \frac{2}{1 \times 2 \times 3} + \frac{2}{2 \times 3 \times 4} + \frac{2}{3 \times 4 \times 5} + \dots + \frac{2}{(k+1)(k+2)(k+3)} = \frac{1}{2} - \frac{1}{(k+2)(k+3)}$	3	There are 4 key parts of the induction; 1. Proving the result true for $n = 1$ 2. Clearly stating the assumption and what is to be proven 3. Using the assumption in the proof 4. Correctly proving the required statement 3 marks • Successfully does all of the 4 key parts 2 marks • Successfully does 3 of the 4 key parts 1 mark • Successfully does 2 of the 4 key parts

Solution		Marks	Comments												
12 (c)...continued. PROOF: $\frac{2}{1 \times 2 \times 3} + \frac{2}{2 \times 3 \times 4} + \frac{2}{3 \times 4 \times 5} + \dots + \frac{2}{(k+1)(k+2)(k+3)}$ $= \frac{2}{1 \times 2 \times 3} + \frac{2}{2 \times 3 \times 4} + \frac{2}{3 \times 4 \times 5} + \dots + \frac{2}{k(k+1)(k+2)} + \frac{2}{(k+1)(k+2)(k+3)}$ $= \frac{1}{2} - \frac{1}{(k+1)(k+2)} + \frac{2}{(k+1)(k+2)(k+3)}$ $= \frac{1}{2} - \frac{(k+3) - 2}{(k+1)(k+2)(k+3)}$ $= \frac{1}{2} - \frac{(k+1)}{(k+1)(k+2)(k+3)}$ $= \frac{1}{2} - \frac{1}{(k+1)(k+2)(k+3)}$ Hence the result is true for $n = k + 1$, if it is true for $n = k$ Since the result is true for $n = 1$, then it is true for all positive integers by induction.															
12 (d) (i) $(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$ Let $x = 2$ $\binom{n}{0} + 2\binom{n}{1} + 4\binom{n}{2} + \dots + 2^n\binom{n}{n}$ $= (1+2)^n$ $= 3^n$		1	1 mark • Correct solution												
12 (d) (ii) $(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$ Differentiating both sides $n(1+x)^{n-1} = \binom{n}{1} + 2\binom{n}{2}x + 3\binom{n}{3}x^2 + \dots + n\binom{n}{n}x^{n-1}$ Let $x = -1$ $\binom{n}{1} - 2\binom{n}{2} + 3\binom{n}{3} - \dots + (-1)^{n-1}n\binom{n}{n}$ $= (-1)(1-1)^{n-1}$ $= 0$		2	2 marks • Correct solution 1 mark • Uses $x = -1$ • Attempts to differentiate both sides of the binomial theorem												
12 (e) (i) $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$ $= x_0 - \frac{e^x - 2x^2}{e^x - 4x}$ $= 1.5 - \frac{e^{1.5} - 2(1.5)^2}{e^{1.5} - 4(1.5)}$ $= 1.487939934\dots$ $= 1.488 \text{ (correct to 3 decimal places)}$		2	2 marks • Correct solution 1 mark • Uses Newton's Method correctly												
12 (e) (ii) <table><tr><td>x</td><td>2.0</td><td>2.1</td><td>2.2</td><td>2.3</td><td>2.4</td></tr><tr><td>$P(x)$</td><td>-0.61</td><td>-0.65</td><td>-0.65</td><td>-0.61</td><td>-0.50</td></tr></table>		x	2.0	2.1	2.2	2.3	2.4	$P(x)$	-0.61	-0.65	-0.65	-0.61	-0.50	1	1 mark • Correctly completed table
x	2.0	2.1	2.2	2.3	2.4										
$P(x)$	-0.61	-0.65	-0.65	-0.61	-0.50										
12 (e) (iii) From the table of values, at $x = 2.3$ the curve is increasing and is on the right hand side of the turning point. The tangent at this point would intersect the x-axis to the right of $x = 2.4$, thus producing an approximation outside the desired interval.		1	1 mark • Valid explanation												

Solution	Marks	Comments
QUESTION 13		
13 (a) (i) $\frac{dV}{dt}$ = rate of water going in – rate of water going out $= \frac{2\pi}{5} - \frac{\pi}{5}\sqrt{h}$ $= \frac{\pi(2 - \sqrt{h})}{5}$	1	1 mark • Correct solution
13 (a) (ii) $V = \pi r^2 h$ $= \pi(2)^2 h$ $= 4\pi h$ $\frac{dV}{dh} = 4\pi$ $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$ $= \frac{1}{4\pi} \times \frac{\pi(2 - \sqrt{h})}{5}$ $= \frac{2 - \sqrt{h}}{20}$	2	2 marks • Correct solution 1 mark • finds $\frac{dV}{dh}$ • uses the chain rule to express $\frac{dh}{dt}$ as a product of other rates
13 (a) (iii) $t = \int_{0.16}^{2.25} \frac{20}{2 - \sqrt{h}} dh$ $= -20 \int_{1.6}^{0.5} \frac{2(2-x)}{2 - (2-x)} dx$ $= 40 \int_{1.6}^{0.5} \frac{2-x}{x} dx$ $= 40 \int_{0.5}^{1.6} \left(\frac{2}{x} - 1 \right) dx$ $= 40 [2 \ln x - x]_{0.5}^{1.6}$ $= 40 \left\{ 2 \ln \left(\frac{1.6}{0.5} \right) - 1.6 + 0.5 \right\}$ $= 49.05206478...$ $\therefore$ it takes 49 minutes to fill the tank $h = (2-x)^2$ $dh = -2(2-x) dx$ when $h = 0.16$ when $h = 2.25$ $0.16 = (2-x)^2$ $2.25 = (2-x)^2$ $2-x = 0.4$ $2-x = 1.5$ $x = 1.6$ $x = 0.5$	3	3 marks • Correct solution using the given substitution <i>Note: solving as an indefinite integral, then using answer to find definite integral is acceptable</i> 2 marks • Correct primitive in terms of x • Correct integrand in terms of x, including the correct limits 1 mark • Correct integrand in terms of x without the limits • Correctly finds answer using an alternative approach
13 (b) (i) $P = 200\,000 - 199\,800e^{-0.01t}$ $\frac{dP}{dt} = 1998e^{-0.01t}$ $= 0.01 \{ 200\,000 - (200\,000 - 199\,800e^{-0.01t}) \}$ $= 0.01(200\,000 - P)$ when $t = 0, P = 200\,000 - 199\,800e^0$ $= 200\,000 - 199\,800$ $= 200$	2	2 marks • Correct solution 1 mark • Establishes initial population is 200 • Verifies given equation is solution to the differential equation
13 (b) (ii) $P > 100\,000$ $200\,000 - 199\,800e^{-0.01t} > 100\,000$ $199\,800e^{-0.01t} < 100\,000$ $e^{-0.01t} < \frac{100\,000}{199\,800}$ $-0.01t < \ln \left(\frac{500}{999} \right)$ $t > 100 \ln \left(\frac{999}{500} \right)$ $t > 69.21466802... \text{ weeks}$ $t > 484.5026762... \text{ days}$ Half of the rabbit population is infected after 485 days	3	3 marks • Correct solution 2 marks • Correct solution, leaving the answer in weeks • Obtains an answer of 483 or 484 days 1 mark • Establishes an inequation, or equation, with t as the subject, using valid methods

Solution	Marks	Comments
13 (c) Join C and E $AB = BE = BC$ (tangents from external point =) so a circle with AC as diameter passes through E $\angle AEC = 90^\circ$ ($\angle$ in semicircle, diameter AC) $\angle CED = 90^\circ$ ($\angle$ in semicircle, diameter CD) $\angle AEC + \angle CED = 180^\circ$ i.e. $\angle AED = 180^\circ$ Thus A, E and D are collinear $90^\circ - \angle CAD + 90^\circ + \angle ECD = 180^\circ$ $\angle ECD = \angle CAD$ OR Join C and E $AB = BE$ (tangents from external point =) $\therefore \angle BAE = \angle AEB$ ($\angle$'s opposite = sides in a Δ are =) $\angle DCA = 90^\circ$ (radius $\perp$ tangent) $\angle DCA + \angle ADC + \angle CAD = 180^\circ$ ($\angle$ sum ΔCAD) $90^\circ + \angle ADC + \angle CAD = 180^\circ$ $\angle ADC = 90^\circ - \angle CAD$ $\angle CED = 90^\circ$ ($\angle$ in semicircle, diameter CD) $\angle EDC + \angle CED + \angle ECD = 180^\circ$ ($\angle$ sum ΔECD) $90^\circ - \angle CAD + 90^\circ + \angle ECD = 180^\circ$ $\angle ECD = \angle CAD$ $\angle FED = \angle ECD$ (alternate segment theorem) Thus $\angle FED = \angle CAD$ $\angle CAD = \angle BAE$ (common $\angle$) $\therefore \angle FED = \angle AEB$ Thus A, D and E are collinear, as the vertically opposite $\angle$'s are equal	4	4 marks • Correct solution 3 marks • Correct solution with poor reasoning • Significant progress towards solution with good reasoning. 2 marks • Significant progress towards solution with poor reasoning. • Progress towards solution with good reasoning. 1 mark • Correctly uses a valid circle geometry theorem.
QUESTION 14		
14 (a) (i) $\dot{x} = V \cos \theta$ $60 = V t \cos \theta$ $\therefore V \cos \theta = 21$ $t = \frac{60}{V \cos \theta}$ $= \frac{60}{21}$ $= 2.857142857...$ $= 2.86 \text{ seconds (to 2 dp)}$	2	2 marks • Correct solution 1 mark • establishes $t = \frac{60}{V \cos \theta}$
14 (a) (ii) Greatest height occurs when $\dot{y} = 0$ and $t = \frac{60}{21}$ $\dot{y} = V \sin \theta - 9.8t$ $0 = V \sin \theta - 9.8 \left(\frac{60}{21} \right)$ $V \sin \theta = 28$ when $t = 0, \dot{y} = V \sin \theta$ $\therefore$ the initial vertical velocity is 28 ms^{-1}	2	2 marks • Correct solution 1 mark • Establishes that the greatest height occurs when $\dot{y} = 0$
14 (a) (iii) Both parts have the same vertical velocity of 0 ms^{-1} at the time of explosion, so $V \sin \theta = 0$ Thus $y = -4.9t^2$ for both parts of the firework i.e. they have the same vertical displacement.	1	1 mark • Correct explanation

Solution		Marks	Comments
14 (a) (iv)	Particle hits the ground when $y = -40$ OR Time to fall = time to rise $-40 = -4.9t^2$ $t^2 = \frac{400}{49}$ $t = \frac{20}{7}$ particle A: $V \cos \theta = 21$ $x_A = 21t$ particle B: $V \cos \theta = \frac{21}{4}$ $x_B = \frac{21}{4}t$ $x_A - x_B = 21t - \frac{21}{4}t$ $= \frac{63}{4}t$ When $t = \frac{20}{7}$; $x_A - x_B = \frac{63}{4} \times \frac{20}{7}$ $= 45$ $\therefore$ the two parts of the firework land 45 metres apart	2	2 marks • Correct solution 1 mark • Finds the time it takes for the two parts of the firework to hit the ground • Establishes that the distance between the two parts of the firework is given by $\frac{63}{4}t$
14 (b) (i)	tangent at P: $y = px - ap^2$ tangent at Q: $y = qx - aq^2$	1	1 mark • Correct answers
14 (b) (ii)	$y = px - ap^2$ $y = qx - aq^2$ $0 = (p - q)x - a(p^2 - q^2)$ $x = \frac{a(p^2 - q^2)}{p - q}$ $x = \frac{a(p - q)(p + q)}{(p - q)}$ $= a(p + q)$ $\therefore R\{a(p + q), apq\}$ $\Rightarrow y = p(a(p + q)) - ap^2$ $= ap^2 + apq - ap^2$ $= apq$	2	2 marks • Correctly shows that R is the point of intersection 1 mark • Correctly finds the x or y value of the point R. • Correctly substitutes R into one of the tangent
14 (b) (iii)	tangents meet the x-axis when $y = 0$ $\therefore A(ap, 0)$ and $B(aq, 0)$ $AB = ap - aq$ $= a(p - q)$ Area = $\frac{1}{2}bh$ $= \frac{1}{2} \times a(p - q) \times (-apq)$ $= \frac{1}{2}a^2pq(q - p)$ Note: as $p > 0$ and $q < 0$ then $apq < 0$	2	2 marks • Correctly shows the area 1 mark • Finds the length of AB • Finds an area using y coordinate of R for the perpendicular height

Solution	Marks	Comments
14(b) (iv)  Drop perpendiculars to the x-axis from P and Q, creating a trapezium $PP'Q'Q$ $\begin{aligned} \text{Area } PP'Q'Q &= \frac{1}{2} \times P'Q' \times (P'P + Q'Q) \\ &= \frac{1}{2} \times (2ap - 2aq) \times (ap^2 + aq^2) \\ &= a^2(p - q)(p^2 + q^2) \end{aligned}$ $\begin{aligned} \text{Area } \triangle OP'P &= \frac{1}{2} \times OP' \times P'P \\ &= \frac{1}{2} \times 2ap \times ap^2 \\ &= a^2p^3 \end{aligned}$ Similarly; replacing p with $-q$ $\text{Area } \triangle OQ'Q = -a^2q^3$ $\begin{aligned} \text{Area } \triangle OPQ &= \text{Area } PP'Q'Q - \text{Area } \triangle OP'P - \text{Area } \triangle OQ'Q \\ &= a^2(p - q)(p^2 + q^2) - a^2p^3 + a^2q^3 \\ &= a^2\{(p - q)(p^2 + q^2) - (p^3 - q^3)\} \\ &= a^2\{(p - q)(p^2 + q^2) - (p - q)(p^2 + pq + q^2)\} \\ &= a^2(p - q)(p^2 + q^2 - p^2 - pq - q^2) \\ &= -a^2pq(p - q) \\ &= a^2pq(q - p) \\ &= 2 \times \text{Area } \triangle ABR \end{aligned}$	3	3 marks • Correct solution 2 marks • Finds the area of $PP'Q'Q$ and the area of $\triangle OP'P$ (or $\triangle OQ'Q$) • Finds the perpendicular distance of O to PQ • Finds the length and equation of PQ 1 mark • Finds the area of $PP'Q'Q$ • Finds the area of $\triangle OP'P$ or $\triangle OQ'Q$ • Finds the length of PQ • Finds the equation of PQ